Capacity Constraints, Random Screening, and Exponential Arrivals

Or Why Journals Must Change

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Abstract

A selection organization faces exponentially increasing arrivals but a fixed capacity to assess submissions substantively. We characterize backlog growth under unrestricted entry, the temporary relief obtained through a fixed preliminary rejection rate, and the rejection probability required to keep substantive assessment within capacity. The central result is that a stable assessment workload can coexist with a probability of substantive review that tends to zero. We distinguish expected load control through independent random rejection from exact capacity control through a randomized admission quota, and address the inherited backlog when screening is introduced after the first year.

1 The capacity-constrained selection problem

Consider years $t = 1, 2, \ldots$. Let A_t be the number of newly arriving items, C the maximum number of substantive assessments per year, and $S \leq C$ the number ultimately selected from those assessments. Suppose arrivals double each year:

$$A_t = A_1 2^{t-1}, \qquad A_1 = 2{,}000, \quad C = 1{,}000, \quad S = 50. \tag{1}$$

Equivalently, $A_t = 1{,}000 \cdot 2^t$. Capacity C is a *flow* (assessments per year), whereas a backlog is a *stock* of work awaiting assessment. We assume that each item requires one assessment, items are processed at most once, and unused annual capacity cannot be carried forward.

Let X_t denote the number of new items permitted into the full-assessment queue after any preliminary screen, D_t the number substantively assessed during year t, and B_t the backlog remaining at year-end. The queue obeys

$$D_t = \min\{C, B_{t-1} + X_t\}, \tag{2}$$

$$B_t = B_{t-1} + X_t - D_t = [B_{t-1} + X_t - C]_+, \qquad B_0 = 0. \tag{3}$$

We first treat admission volumes deterministically (a *fluid approximation*) and then examine genuinely random rejection. A constant arrival rate *within* each year does not change the annual stock accounting; it can affect individual waiting times.

*This paper is written as an allegory for the purpose of explaining why journals must change in our new academic research world of LLMs and other AI tools. GPT-6 and I wrote this paper in 15 minutes.

This working paper was generated primarily with the assistance of generative AI. AI systems contributed substantially to the development of the arguments and the drafting of the manuscript. It is presented as an exploratory theoretical work, not as a peer-reviewed publication. Readers should independently evaluate its reasoning, references, and conclusions.

2 Unrestricted admission and backlog growth

Without screening, $X_t = A_t$. Since $A_1 > C$, capacity is exhausted in every year and the queue never empties.

Proposition 1 (Exponential backlog under fixed capacity)**.** *Under* (1)*, with every arrival admitted for assessment and* $B_0 = 0$*, the end-of-year backlog is*

$$B_t = \sum_{j=1}^{t} A_j - tC = 1{,}000(2^{t+1} - 2 - t). \tag{4}$$

In particular, B_t *grows at rate* $\Theta(2^t)$*.*

Proof. The cumulative arrivals through year t equal $A_1(2^t - 1) = 1{,}000(2^{t+1} - 2)$. The queue always contains enough items for C assessments per year. Subtracting the cumulative tC assessments gives (4); its exponential term dominates the linear term. □

The first five end-of-year backlogs are 1,000, 4,000, 11,000, 26,000, and 57,000. At the end of Year 5, even an immediate cessation of new arrivals would leave 57 years' worth of work at the prevailing assessment rate.

3 Fixed-rate screening: temporary relief

Suppose screening begins in Year 2, after Year 1 has generated $B_1 = 1{,}000$ unassessed items. From Year 2 onward, an initial screen rejects 80% of new arrivals and admits the other 20% for full assessment. In the fluid approximation,

$$X_t = 0.2A_t, \qquad t \geq 2. \tag{5}$$

In Year 2, only 800 items enter the queue, but the inherited backlog remains. The end-of-year backlog falls to 800. As the admitted flow subsequently doubles, it once again outstrips capacity. Specifically, for $t \geq 2$,

$$B_t = 1{,}000 + \sum_{j=2}^{t}(0.2A_j - 1{,}000) = 200 \cdot 2^{t+1} + 1{,}200 - 1{,}000t. \tag{6}$$

For the stated parameters, the right-hand side stays nonnegative, so the queue never becomes empty under this approximation. An 80% screen reduces the Year 5 backlog from 57,000 to 9,000, but it does not alter its long-run exponential order.

4 Endogenous rejection: preserving a fixed assessment workload

Let p_t be the probability that a new arrival is rejected at the preliminary stage, with independent draws across arrivals. The mean number admitted is

$$\mathbb{E}[X_t] = (1 - p_t)A_t. \tag{7}$$

If no backlog is inherited and the policy seeks to admit *on average* exactly C items in year t, it must set

$$\boxed{p_t^* = 1 - \frac{C}{A_t} = 1 - 2^{-t}.} \tag{8}$$

This expression is meaningful when $A_t \geq C$, as in the present example.

Proposition 2 (Asymptotic exclusion)**.** *With $A_t = A_1 2^{t-1}$ and constant $C > 0$, the required rejection probability p_t^* in (8) increases monotonically toward one. The probability of reaching substantive review, $1 - p_t^* = C/A_t$, falls by one-half each year.*

Proof. Because $C/A_t = (C/A_1)2^{-(t-1)}$, the admission fraction is positive, strictly decreasing, and converges to zero. Therefore its complement increases toward one. □

Table 1: Arrival growth and the no-new-backlog screening rule

Year	New arrivals A_t	Admitted quota	Rejection p_t^*	Mean selection rate[†]
1	2,000	1,000	50.000%	2.50000%
2	4,000	1,000	75.000%	1.25000%
3	8,000	1,000	87.500%	0.62500%
4	16,000	1,000	93.750%	0.31250%
5	32,000	1,000	96.875%	0.15625%
6	64,000	1,000	98.4375%	0.078125%
7	128,000	1,000	99.21875%	0.0390625%
8	256,000	1,000	99.609375%	0.01953125%

[†]Ex ante probability for an exchangeable arrival when exactly $S = 50$ items are selected each year and the annual admission quota is $C = 1{,}000$. These figures refer to screening from Year 1, not to the inherited-backlog scenario.

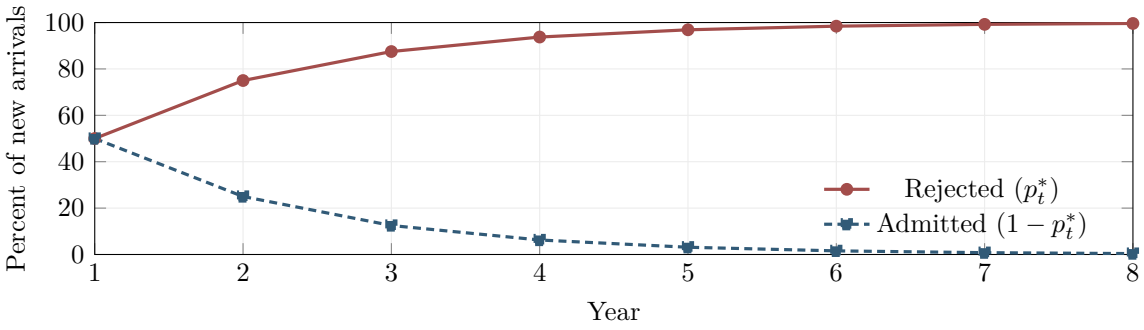


Figure 1: Endogenous screening with fixed capacity and annual doubling. The substantive assessment workload can remain unchanged while access to it collapses.

4.1 An inherited backlog changes the rule

The path in (8) assumes there is *no inherited queue*. If screening first begins in Year 2, with $B_1 = 1{,}000$, assigning another 1,000 new items to assessment in Year 2 would leave the old 1,000-item backlog intact. Clearing that backlog *by the end of Year 2* requires admitting zero new arrivals that year, using all capacity to assess the inherited stock. Thus $p_2 = 1$ (all 4,000 new arrivals are rejected). From Year 3 onward, $p_t = 1 - 2^{-t}$ again prevents a new queue from forming.

More generally, if $0 \le B_{t-1} \le C$, the maximum number of new admissions consistent with *clearing the queue by year-end* is $C - B_{t-1}$. Its corresponding mean-load rejection rule is

$$\boxed{p_t = 1 - \frac{C - B_{t-1}}{A_t}.} \tag{9}$$

If $B_{t-1} > C$, even rejecting every new arrival cannot eliminate the backlog in that year.

5 Why a probability does not guarantee an empty queue

There is an important distinction between *expected admissions* and *certain capacity compliance*. Under independent random screening,

$$X_t \sim \text{Binomial}(A_t, 1 - p_t), \qquad \text{Var}(X_t) = A_t p_t (1 - p_t). \tag{10}$$

At $p_t = p_t^*$, $\mathbb{E}[X_t] = C$ but X_t can exceed C.

Proposition 3 (Mean-load matching does not ensure zero backlog)**.** *Suppose* $B_{t-1} = 0$, $A_t > C$, *and* $0 < p_t^* < 1$. *Independent screening with* $p_t = p_t^*$ *gives* $\mathbb{E}[X_t] = C$, *yet*

$$\mathbb{P}(B_t > 0) > 0, \qquad \mathbb{E}[B_t] > 0. \tag{11}$$

Proof. The binomial variable X_t has positive probability of taking values above its mean C (in particular, all A_t items can be admitted, with positive probability). On those outcomes, $B_t = (X_t - C)_+ > 0$; hence the positive-part random variable has strictly positive expectation. □

A rule that actually *guarantees* zero backlog, starting from an empty queue, is to draw *exactly* C of the A_t arrivals uniformly without replacement for full assessment. Each item's inclusion probability is C/A_t, and the implied rejection probability is still p_t^*, but the number of admissions is no longer binomial: it equals C with certainty. In the presence of a queue $B_{t-1} \leq C$, the organization instead draws exactly $C - B_{t-1}$ new items for admission, conditional on the inherited stock being assessed that year. This is a randomized *quota*, not independent Bernoulli screening.

6 Comparison over the first five years

Table 2: End-of-year backlog across three regimes (items)

Year	Arrivals	No screening	Fixed 80% screen[‡]	Adaptive quota[§]
1	2,000	1,000	1,000	1,000
2	4,000	4,000	800	0
3	8,000	11,000	1,400	0
4	16,000	26,000	3,600	0
5	32,000	57,000	9,000	0

[‡]Screening begins in Year 2; figures use deterministic admitted volumes equal to 20% of arrivals. [§]No screen in Year 1; Year 2 rejects all new arrivals to clear the inherited queue; from Year 3, a random quota admits exactly 1,000 per year.

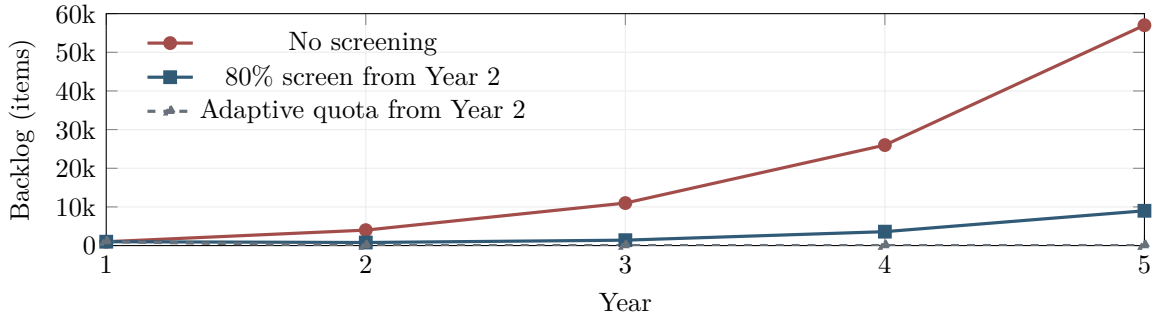


Figure 2: A fixed rejection probability delays congestion; an appropriately sized quota eliminates the assessment backlog by excluding an increasing proportion of new arrivals.

7 Selection probabilities and institutional interpretation

Provided a queue-free system assesses exactly C items and selects exactly S in each year, the *average* conditional selection rate among assessed items is

$$q = \frac{S}{C} = \frac{50}{1{,}000} = 0.05. \tag{12}$$

Under an exchangeability assumption for new arrivals (and uniform random admission under a quota), the ex ante selection probability of a newly arriving item is

$$\mathbb{P}(\text{selection in year } t) = \underbrace{\frac{C}{A_t}}_{\text{admitted}} \underbrace{\frac{S}{C}}_{\text{selected} \mid \text{assessed}} = \frac{S}{A_t} = 0.05 \cdot 2^{-t}. \tag{13}$$

The numerical sequence starts at 2.5% in Year 1 and falls to 0.15625% in Year 5. If merit or other characteristics predict selection, individual probabilities will vary; S/A_t then represents a cohort average, not a universal individual probability. Further, (13) applies to the queue-free regime, not to a transitional year in which assessment capacity is devoted to inherited submissions.

Corollary 1 (Stable organizational output, vanishing individual access)**.** *If A_t doubles indefinitely, C and S remain fixed, and each year's assessment admissions are quota-limited to C, annual substantive assessments and final selections can remain constant at C and S, while the average ex ante probability that an arrival receives substantive review, or is selected, converges to zero.*

Proof. Quota-limited annual admissions and selections are constants by construction, while $C/A_t \to 0$ and $S/A_t \to 0$ as $t \to \infty$. □

The result separates *administrative throughput* from *access to institutional judgment*. An organization may report stable processing times, a cleared assessment queue, and unchanged annual selections, even as a rapidly increasing fraction of arrivals is excluded before substantive consideration. Random screening distributes access symmetrically in a formal sense, but it is not a substitute for evaluation: eligible or high-quality items may be rejected alongside weak ones.

The model deliberately treats the preliminary screening stage as sufficiently scalable that it can handle all A_t arrivals without using the fixed full-assessment capacity. This is a significant assumption. If screening itself requires nontrivial human labor or computational expenditure per item, its aggregate resource requirements also increase exponentially, and the apparent stability of the second-stage queue need not imply stability of the entire system.

8 Conclusion

Exponential growth in arrivals can coexist with a stable assessment workload only if access is rationed increasingly severely. Eliminating the queue does not expand substantive evaluative capacity.